

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I D.HUSSAINIAH(192425300) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D.HUSSAINIAH

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	25%	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	25%	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	20%	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	15%	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Answer:

An image appears too bright when most of its pixels have **high intensity values**. Because of excessive brightness, details in bright areas may be lost or difficult to distinguish. A suitable grey-level transformation can be used to reduce the intensity and improve visibility.

A grey-level transformation changes the intensity of each pixel according to a transformation function:

$$\begin{bmatrix} s = T(r) \end{bmatrix}$$

where (r) is the input intensity and (s) is the output intensity.

For an excessively bright image, **intensity reduction or gray-level scaling** can be applied. The high-intensity pixels are mapped to lower or more suitable intensity values.

Working:

1. Take the bright image as input.
2. Examine the intensity values of the pixels.
3. Apply the selected grey-level transformation to reduce excessive intensity.
4. Map the transformed values into the required grey-level range.
5. Generate the enhanced image.

For example, if a pixel has a very high intensity value, the transformation reduces its intensity so that details in that region become more visible.

Expected Result:

- Excessive brightness is reduced.
- Details in bright regions become clearer.
- The intensity range becomes more suitable for viewing.
- The overall image appears more balanced.

Justification:

Since the main problem is **excessively high intensity**, reducing the intensity values is appropriate. It improves visibility without requiring complex image processing.

2. A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Answer:

A dark image contains most of its pixels in the **low-intensity range**. Important objects and details may therefore be difficult to see. A suitable transformation for this situation is the **logarithmic transformation**.

The logarithmic transformation is represented by:

$$\begin{bmatrix} s = c \log(1+r) \end{bmatrix}$$

where:

- (r) = input pixel intensity
- (s) = output intensity
- (c) = positive scaling constant

Working:

The logarithmic transformation gives greater expansion to low-intensity values. Therefore, small differences between dark pixels become more visible.

The process is:

1. Take the dark image as input.
2. Obtain the intensity value of each pixel.
3. Apply the logarithmic transformation.
4. Scale the output values to the required grey-level range.
5. Produce the enhanced image.

Example:

In a night-time photograph, an object may have intensity values very close to black. Log transformation increases the visibility of these low-intensity values, making the object easier to identify.

Expected Result:

- Dark areas become brighter.
- Details hidden in dark regions become visible.
- Low-intensity differences are expanded.
- Overall image visibility improves.

Justification:

Logarithmic transformation is suitable because it specifically **expands low-intensity regions**, which is the main requirement for improving a dark image.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Answer:

An image has poor contrast when its pixel intensities are concentrated within a **small or narrow range**. As a result, objects and background may have similar intensity values. **Histogram Equalization** is a suitable histogram-based technique for improving such an image.

A histogram represents the number of pixels present at each intensity level. Histogram equalization redistributes the intensity values over a wider range.

The transformation is based on the cumulative distribution function (CDF):

$$\begin{bmatrix} s = (L-1) \text{CDF}(r) \end{bmatrix}$$

where (L) represents the number of grey levels.

Working:

1. Calculate the histogram of the input image.
2. Find the probability of occurrence of each intensity.
3. Calculate the cumulative distribution function.
4. Use the CDF to obtain new intensity values.
5. Replace the original intensity values with the transformed values.

Example:

If most pixels of an image are concentrated between intensity values 80 and 150, the image may look dull. Histogram equalization spreads the intensities over a larger available range.

Expected Result:

- Overall contrast increases.
- Dark and bright regions become more distinguishable.
- Important image details become clearer.
- The available intensity range is used more effectively.

Justification:

Since the problem is **poor contrast caused by an ineffective intensity distribution**, histogram equalization is suitable because it directly modifies the histogram and redistributes the intensity values.

4.A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Answer:

For a low-contrast image that requires a better distribution of intensity levels, **Histogram Equalization** is an appropriate technique. It improves the image by redistributing the pixel intensities across a wider range.

Concept:

Histogram equalization uses the image histogram and its **cumulative distribution function (CDF)** to create a transformation between the original and new intensity values.

$$\begin{bmatrix} s = (L-1) \text{CDF}(r) \end{bmatrix}$$

Working Process:

1. Obtain the histogram of the image.
2. Calculate the frequency of each intensity level.
3. Normalize the histogram to obtain probabilities.
4. Calculate the cumulative distribution.
5. Map the original intensity values to new values.

Effect on Intensity Levels:

Before equalization, many pixels may be concentrated in a narrow range. This produces low contrast.

After equalization, the intensity values are spread over a wider range. This increases the difference between different regions of the image.

Expected Result:

- Better intensity distribution.
- Improved contrast.
- Dark and bright regions become easier to distinguish.
- Details become more visible.

Justification:

The method is suitable because the question specifically requires improvement in the **distribution of intensity levels**, which is the main purpose of histogram equalization.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Answer:

Uneven brightness occurs when different regions of an image receive different amounts of illumination. For example, one side of a document may be bright while the other side is dark. **Homomorphic filtering** is an appropriate technique for correcting non-uniform illumination.

An image can be represented as:

$$\begin{bmatrix} f(x,y) = i(x,y)r(x,y) \end{bmatrix}$$

where:

- $(i(x,y))$ represents the illumination component.
- $(r(x,y))$ represents the reflectance component.

The illumination usually changes slowly across an image, while reflectance contains important details and edges.

Working:

1. Take the input image.
2. Apply logarithmic transformation.
3. Convert the image into the frequency domain using Fourier Transform.
4. Reduce the low-frequency illumination component using filtering.
5. Preserve or enhance useful high-frequency details.
6. Apply inverse Fourier Transform to obtain the enhanced image.

Example:

In a scanned document, the top portion may be brighter than the bottom portion. Homomorphic filtering reduces this uneven illumination and makes the background more uniform.

Expected Result:

- Uneven brightness is reduced.
- Background illumination becomes more uniform.
- Details and edges become clearer.
- Contrast is improved.

Justification:

Simple brightness adjustment cannot effectively handle brightness that changes from one region to another. Homomorphic filtering is suitable because it separates illumination from image reflectance.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Answer:

Noise produces unwanted and random variations in pixel intensity. A **mean filter**, also called an averaging filter, can be used to reduce random intensity variations.

The mean filter replaces each pixel with the average of the pixels in its neighborhood.

For a (3×3) neighborhood:

$$g(x,y)=\frac{1}{9}\sum_{i=-1}^1\sum_{j=-1}^1f(x+i,y+j)$$

Working:

1. Select a small neighborhood such as (3×3).
2. Consider all pixels inside the neighborhood.
3. Add their intensity values.
4. Calculate their average.
5. Replace the centre pixel with this average.
6. Move the filter across the entire image.

Example:

If one pixel has an unusually high value because of noise while the surrounding pixels have similar lower values, averaging the neighborhood reduces the effect of that noisy pixel.

Effect:

- Random intensity variations are reduced.
- The image becomes smoother.
- Small fluctuations are suppressed.
- However, excessive smoothing may blur edges.

Justification:

The mean filter is suitable for general random variations because averaging neighboring pixels reduces sudden intensity changes caused by noise.

Expected Result:

The output image contains less random noise and is smoother, making it more suitable for further image processing.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Answer:

When noise needs to be removed while preserving important image details, a **median filter** is an appropriate choice. It is particularly effective for **salt-and-pepper noise**.

Unlike the mean filter, the median filter does not calculate an average. It sorts the values in a neighborhood and selects the middle value.

Working:

1. Select a neighborhood, such as (3x3).
2. Collect all pixel values in that neighborhood.
3. Arrange the values in ascending order.
4. Select the middle value.
5. Replace the centre pixel with the median.
6. Repeat this process throughout the image.

Example:

Consider the values:

```
[  
10,11,12,13,255,14,12,11,10  
]
```

After sorting:

```
[  
10,10,11,11,12,12,13,14,255  
]
```

The median is 12. Therefore, the extreme noisy value 255 can be replaced by 12.

Why It Preserves Details:

The median filter removes extreme values without averaging across neighbouring edges. Therefore, important boundaries are generally preserved better than with simple averaging.

Expected Result:

- Noise is reduced.
- Salt-and-pepper noise is removed.
- Edges are better preserved.
- Important details remain relatively clear.

Justification:

The median filter provides a good balance between **noise removal and detail preservation**, making it suitable when maintaining image structure is important.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Answer:

Smoothing filters reduce noise, but they can also remove high-frequency components such as edges and fine details. This can make an image appear blurred. **Laplacian sharpening** can be used to restore the lost sharpness.

The Laplacian is a second-order derivative operator that detects rapid changes in intensity.

A common Laplacian mask is:

```
[  
\begin{bmatrix}  
0&-1&0\\  
-1&4&-1\\  
0&-1&0  
\end{bmatrix}  
]
```

Working:

1. Take the blurred image.
2. Apply the Laplacian mask to the image.
3. The filter detects edges and rapid intensity changes.
4. Combine the Laplacian information with the original image.
5. Produce the sharpened image.

A sharpening operation can be represented as:

$$\begin{bmatrix} g(x,y)=f(x,y)-\nabla^2 f(x,y) \end{bmatrix}$$

depending on the selected Laplacian convention.

Effect:

The Laplacian emphasizes high-frequency components such as:

- Edges
- Fine structures
- Object boundaries

Expected Result:

- Blurred edges become sharper.
- Fine details become more visible.
- Object boundaries become clearer.
- Overall image sharpness improves.

Justification:

Since smoothing has reduced high-frequency information, Laplacian sharpening is suitable because it detects and enhances these high-frequency details.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Answer:

Fine details, edges, and textures are mainly represented by **high-frequency components** of an image. Therefore, **high-pass filtering** can be used to enhance these details.

A high-pass filter allows high-frequency components to pass while reducing low-frequency components.

Working Principle:

1. Take the input image.
2. Transform the image into the frequency domain using Fourier Transform.
3. Apply a high-pass filter.
4. Low-frequency components are reduced.
5. High-frequency components are preserved or enhanced.
6. Apply inverse Fourier Transform to obtain the enhanced image.

The frequency-domain operation can be represented as:

$$\begin{bmatrix} G(u,v) = H(u,v)F(u,v) \end{bmatrix}$$

where $(H(u,v))$ is the high-pass filter and $(F(u,v))$ is the Fourier transform of the image.

Effect:

High-frequency components correspond mainly to:

- Edges
- Fine details
- Textures
- Rapid intensity changes

Therefore, enhancing them makes these structures more visible.

Example:

A fingerprint image contains thin ridge patterns. High-pass filtering can enhance these patterns and make them easier to identify.

Expected Result:

- Fine details become clearer.
- Edges become more prominent.
- Texture information is enhanced.
- Image features become easier to analyze.

Justification:

Because the question specifically focuses on **fine details**, high-pass filtering is suitable since fine details are associated with high-frequency components.

10. A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Answer:

Noise should generally be reduced before performing further image processing because it can produce incorrect edges, segmentation errors, and false features. A **median spatial filter** is a suitable preprocessing method, especially for salt-and-pepper noise.

Concept:

Spatial filtering operates directly on the pixels of an image using a local neighborhood.

For a median filter:

$$[\\ g(x,y)=\\text{median}\\{f(s,t):(s,t)\\in S_{xy}\\}\\]$$

Working:

1. Take the noisy image as input.
2. Select a small window, such as $(3 \\times 3)$.
3. Collect the pixel values inside the window.
4. Sort the values.
5. Select the middle value.
6. Replace the centre pixel with the median.
7. Repeat the operation over the complete image.

Example:

For a neighborhood containing:

$$[\\ 20,21,20,21,255,22,20,21,22 \\]$$

the value 255 is an extreme noisy value. The median is close to the surrounding values, so the noise is removed.

Importance Before Further Analysis:

Noise reduction is useful before:

- Edge detection
- Segmentation
- Feature extraction
- Object recognition

It reduces false features and improves the reliability of later processing.

Expected Result:

- Noise is reduced.
- Important edges are relatively preserved.
- Image quality improves.
- The image becomes better suited for further computer vision analysis.

Justification:

Median filtering is suitable because it directly processes neighboring pixels and effectively removes isolated extreme intensity values while preserving important image boundaries.